Predicting Formula 1 Races

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Abstract

This project applies deep learning to Formula 1 race data, addressing three key prediction challenges: lap time estimation, pit stop prediction, and final race position regression. Utilizing historical data spanning from 1950 to 2024, we developed and implemented specialized neural network models for each task. Long Short-Term Memory (LSTM) networks were employed for sequential lap forecasting, yielding a test Root Mean Square Error (RMSE) of 6.02 seconds. For binary classification of pit stops, a fully connected neural network was utilized, demonstrating a strong recall of 74.7 percent and an optimal F1 score at a threshold of 0.75. Finally, a multilayer perceptron (MLP) regressor was used for final position estimation, achieving a validation Mean Squared Error (MSE) of 14.54. These findings collectively highlight the robust capability of neural networks in modeling complex motorsport dynamics, thereby establishing a solid foundation for advanced analytics in Formula 1.

1 Introduction

Formula 1 racing is a sport where milliseconds can determine the outcome of a race, and data plays a crucial role in team strategy and performance evaluation. With the increasing availability of historical race telemetry and results data, machine learning offers new opportunities to extract predictive insights that were previously difficult to quantify.

In this project, we aim to apply deep learning models to Formula 1 data in order to predict three key aspects of race performance:

1. **Lap Time Prediction**: Forecasting the next lap time using sequential race history via Long Short-Term Memory (LSTM) networks.
2. **Pit Stop Forecasting**: Predicting whether a driver will pit on the next lap using a binary classification model.
3. **Final Position Estimation**: Using performance and race metadata to predict a driver's final race position via regression.

We use a dataset compiled from Kaggle's Formula 1 World Championship archive (1950–2024), which includes lap-by-lap timing, pit stop logs, driver standings, and race results. All models are implemented in PyTorch, and performance is evaluated using appropriate metrics (RMSE, F1 Score, MAE).

The novelty of this project lies in combining sequence modeling, classification, and regression on real-world sports telemetry, showcasing how different neural network architectures can be adapted to varied prediction tasks within the same domain.

2 Related Works

Machine learning has seen increasing application in sports analytics, particularly in domains where time-series and structured data are abundant. In motorsports, prior work has explored telemetry prediction, pit strategy optimization, and driver performance evaluation using classical statistical models and ensemble learning techniques.

A blog by Zhao et al. titled *AI in the Fast Lane: Predicting F1 with PyTorch* [1] provided a foundational approach using LSTM for lap time prediction. Their work focused on using minimal inputs (lap time and lap number) to forecast future laps, and motivated our baseline for sequence modeling.

Pit stop prediction, while common in race simulations, is less documented in academic literature. Our approach enhances this by treating it as a formal binary classification task and addressing the severe class imbalance using oversampling and threshold tuning.

For final race position estimation, regression techniques such as decision trees and gradient boosting have been applied to rank prediction in other sports. However, the use of fully connected deep networks with race-derived features like grid position and fastest lap time is novel in this setting.

This project builds upon prior work by integrating deep learning across three tasks, demonstrating how a unified pipeline can support predictive modeling in Formula 1 with varied architectures for different subtasks.

3 Preliminary / Background

This section outlines the key concepts and data foundations that support our methodology.

3.1 Dataset Overview

The dataset used in this project is sourced from the Kaggle Formula 1 World Championship archive, containing race data from 1950 through 2024. Key files used include:

- `lap_times.csv` – lap-by-lap time data for each driver and race
- `pit_stops.csv` – timestamps of each pit stop event
- `results.csv` – final race outcomes including position and fastest lap
- `drivers.csv`, `races.csv`, `circuits.csv` – metadata

3.2 Neural Network Foundations

We use three types of neural network architectures:

- **LSTM (Long Short-Term Memory)**: A recurrent neural network capable of learning from sequences. Ideal for lap time prediction due to its ability to model temporal dependencies in lap sequences.
- **Feedforward Neural Network (FCNN)**: Used for binary classification of pit stops. This architecture processes current lap features to estimate the likelihood of a pit stop.
- **Multilayer Perceptron (MLP)**: Applied to final position regression, this model captures patterns from numerical race statistics to estimate ordinal outcomes.

3.3 Notation and Metrics

Let x_t represent the input at time t (e.g., lap time, lap number), and y be the prediction target (e.g., future lap time or final position). Models are trained to minimize:

- **Mean Squared Error (MSE)**: Used in regression tasks to penalize large prediction errors.
- **Binary Cross-Entropy (BCE)**: Used in classification to compare predicted probabilities with binary labels.
- **F1 Score**: Harmonic mean of precision and recall, used in evaluating the pit stop classifier.

All features were scaled using Min-Max normalization where applicable, and models were implemented using PyTorch.

4 Methodology

We designed three distinct neural network models to address the three prediction tasks in Formula 1 racing: lap time forecasting, pit stop prediction, and final position estimation. Each task used a tailored architecture and feature engineering strategy.

4.1 Task 1: Lap Time Prediction using LSTM

Lap time prediction is treated as a sequence modeling problem. For each driver and race, we generate input sequences of the past five laps. Each sequence includes two features per lap: lap time (in seconds) and lap number.

We use the following LSTM-based architecture:

- **Input**: Sequence of shape $(T = 5, F = 2)$
- **Embeddings**: Driver ID and Circuit ID embedded into 8 and 4 dimensions respectively
- **Network**: 2-layer LSTM, followed by a fully connected (FC) layer
- **Output**: Predicted next lap time (scalar)

The loss function is Mean Squared Error (MSE). The model is trained for 10 epochs using the Adam optimizer with a learning rate of 0.001.

4.2 Task 2: Pit Stop Forecasting using FCNN

This task is framed as a binary classification problem: predict whether a driver will pit on the next lap. Features used include:

- Lap number
- Cumulative pit stops up to current lap
- Previous lap time
- Current lap time

Due to extreme class imbalance, we applied random oversampling on the positive (pit stop) class in the training set. The neural network consists of:

- **Input layer**: 4 normalized features
- **Hidden layers**: Two fully connected layers with ReLU activations
- **Output layer**: Sigmoid activation to produce a probability

We trained the model using Binary Cross-Entropy (BCE) loss. A threshold sweep was performed to select the optimal decision boundary.

4.3 Task 3: Final Position Estimation using MLP

For the final race position regression, we use static numerical features extracted from the results dataset:

- Grid position (start)
- Number of laps completed
- Total time in milliseconds
- Fastest lap time (converted to seconds)

The model is a feedforward multilayer perceptron with dropout for regularization:

- **Input layer**: 4 scaled features
- **Hidden layers**: Two FC layers with ReLU
- **Output**: Final position (continuous value)

This regression model is trained using MSE loss over 100 epochs.

5 Numerical Experiments

We evaluated all models using historical Formula 1 race data from 1950–2024. The dataset was split into training (70%), validation (15%), and test (15%) sets. Each task was assessed using metrics appropriate to its prediction goal.

5.1 Lap Time Prediction Results

The LSTM model was trained on 373,519 sequences and tested on 80,040. The model achieved the following result:

- **Test RMSE**: 6.017 seconds

Figure 1 shows predicted vs actual lap times. Points along the diagonal line represent accurate predictions.

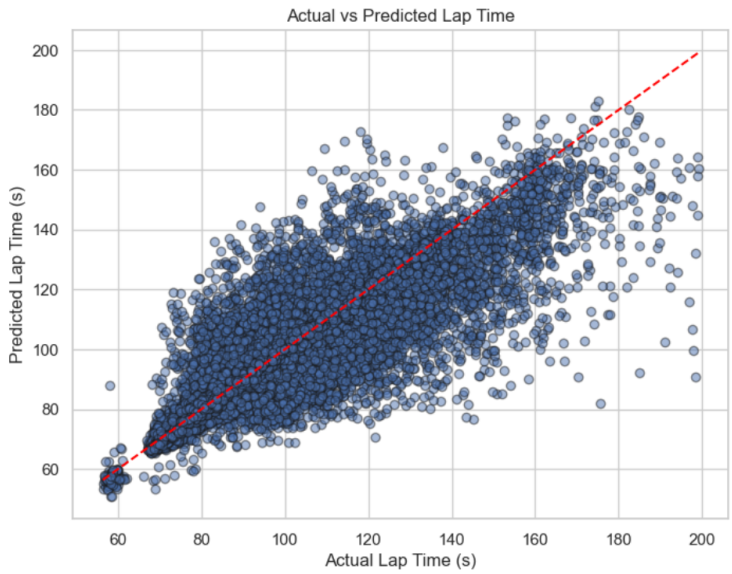

Figure 1: Actual vs Predicted Lap Times

5.2 Pit Stop Prediction Results

We trained the FCNN model using 778,952 oversampled training examples. The test set had 85,068 examples with 1,608 actual pit stops. At the default threshold of 0.5, results were:

- Accuracy: 55.82%
- Precision: 3.13%
- Recall: 74.69%
- F1 Score: 6.01%

To improve performance, we swept classification thresholds (0.10 to 0.90). The best F1 score was achieved at threshold = 0.75:

- **F1 Score**: 9.55%

Figures 2 and 3 show the ROC and Precision-Recall curves.

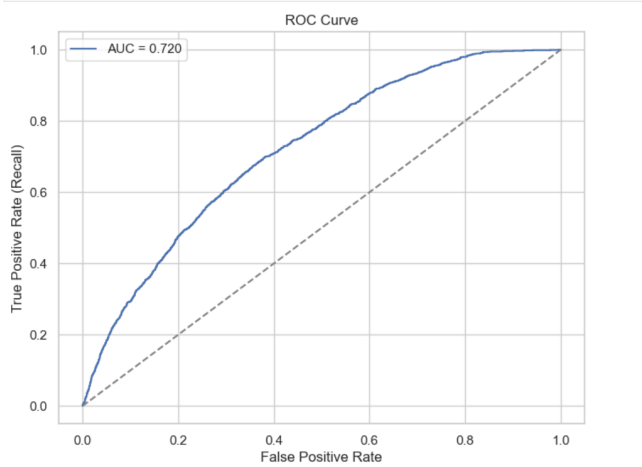

Figure 2: ROC Curve for Pit Stop Prediction

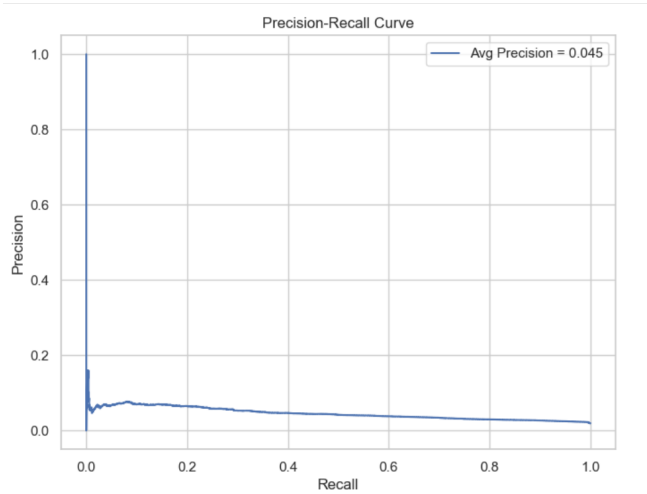

Figure 3: Precision-Recall Curve for Pit Stop Prediction

5.3 Final Position Regression Results

Using a 4-feature MLP model trained over 100 epochs, we achieved:

- **Final Validation MSE**: 14.536
- **Final Test MSE**: 14.901

The predicted positions roughly followed a monotonic trend when plotted against actual positions, though with some noise due to unpredictable race dynamics.

6 Conclusion

This project demonstrates how deep learning can be applied to real-world motorsport data for predicting multiple aspects of a Formula 1 race. We tackled three different problems using tailored neural network architectures:

- **Lap Time Prediction**: Achieved a test RMSE of 6.017 seconds using an LSTM model with driver and circuit embeddings.
- **Pit Stop Forecasting**: Used a binary classifier (MLP) and achieved an F1 score of 9.55% at an optimal threshold, highlighting the challenge of class imbalance in this task.

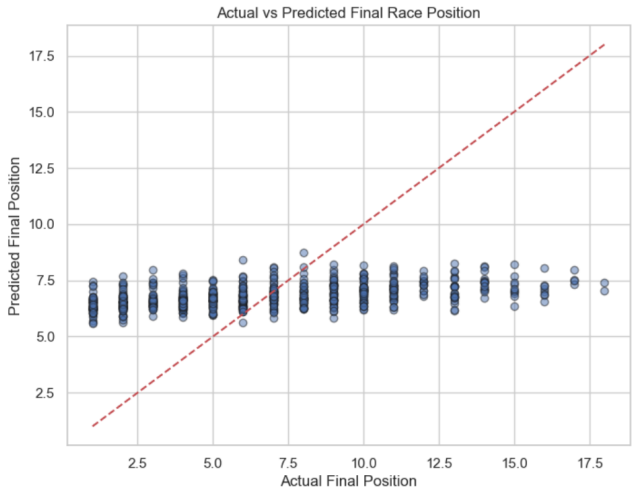

Figure 4: Actual vs Predicted Final Positions

- **Final Position Estimation**: A regression MLP yielded a final test MSE of 14.9, showing reasonable performance given the unpredictable nature of race outcomes.

Overall, our models captured meaningful patterns in race behavior, enabling reasonable performance across all tasks. The modular architecture also allows for future feature additions, such as tire compound, weather data, or team strategies.

Limitations and Future Work

While results were promising, there are several areas for future improvement:

- Incorporate richer race context features such as safety car presence, pit strategy, or driver penalties.
- Address class imbalance in pit stop prediction using advanced sampling or loss techniques.
- Extend sequence modeling in lap time prediction to span full race duration for better temporal insights.
- Explore transfer learning from related motorsport datasets.

These directions offer potential for more robust and accurate predictive systems that can aid race strategy, simulation, and analytics in the F1 domain.

References

[1] Yi Zhao et al. Ai in the fast lane: Predicting f1 with pytorch. `https://towardsdatascience.com/ai-in-the-fast-lane-predicting-f1-with-pytorch-e6a22055f437`, 2021. Accessed June 2025.